

A new sensitive chemical actinometer

II. Potassium ferrioxalate as a standard chemical actinometer

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The earlier experiments with potassium ferrioxalate have been extended and a detailed study has been made of the photolysis of the acidified solutions previously recommended for chemical actinometry (Parker 1953). Accurate values of quantum efficiency have been determined at twelve wavelengths between 254 and 578 $m\mu$, and the effect of temperature, light intensity and of photolyte composition have been investigated. At some wavelengths independent values of quantum efficiency have been obtained by comparison with uranyl oxalate and by direct comparison with a calibrated thermopile. The results indicate that potassium ferrioxalate provides a very valuable new chemical actinometer, and its use is recommended in place of uranyl oxalate for many purposes. Full working details for the use of the actinometer are provided.

INTRODUCTION

Since the classic investigations of uranyl oxalate (Leighton & Forbes 1930; Heidt & Daniels 1932; Brackett & Forbes 1933; Forbes & Heidt 1934), this substance has held the field as a chemical actinometer, although it suffers from some disadvantages. In principle a chemical method has some important advantages over physical methods of light measurement and, for general use, its main requirements are constant quantum efficiency and high absorption factor over a wide range of wavelengths and of total radiation dose, high sensitivity and precision, coupled with simplicity of operation and ready availability of the photochemical material. Experiments with solutions of potassium ferrioxalate had shown that this substance to a large degree fulfils these conditions (Parker 1953). Since the publication of this preliminary work, we have made a detailed study of the characteristics of this system with the object of establishing it as a standard actinometer for use both in photochemical investigations and for other purposes. During this period we have also made extensive use of the actinometer for the measurement of monochromatic and polychromatic light, in some cases at very low intensities, and our experience has convinced us of its value. The results of our investigation, together with working instructions for using the actinometer, are described in this paper.

QUANTUM EFFICIENCY BY REFERENCE TO URANYL OXALATE

The results of comparison with the uranyl oxalate system at 366 and 254 $m\mu$ were reported in part I (Parker 1953). The data have now been extended to include 405 and 436 $m\mu$ using filtered light from a 125 W medium-pressure mercury lamp and 334, 313 and 302 $m\mu$ using light from a 1 kW high-pressure mercury lamp isolated by means of a Hilger D 96 large-aperture quartz monochromator (see below). Additional values at 366 and 254 $m\mu$ have also been obtained.

For the determination at 405 and 436 $m\mu$ (and also for the determinations at 546 and 578 $m\mu$; see below) the light from the 125 W mercury lamp (type MB/D, operated from 50 c/s a.c. mains) was roughly collimated by lenses and passed through a 2 cm depth of 10 % w/v $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ solution and then through a double-compartment optical cell (1 and 2 cm optical depth) containing appropriate filter solutions (see table 1), or through glass filters. The filtered beam was focused on a rectangular aperture (2×1.5 cm), behind which was situated the actinometer cell (35 by 45 mm and 15 mm optical depth) containing 15 ml. of photolyte. Between the last lens and the aperture was placed a clear glass plate at an angle of 45° which served to deflect a proportion of the light on to the phototube monitoring unit (Parker 1953). The actinometer cell was seated in a hollow brass thermostatically controlled block which fitted closely round the bottom and two sides of the cell and the temperature of the photolyte could be maintained constant to within $\pm 0.2^\circ \text{C}$.

At each wavelength the purity of the filtered light was checked by means of a Hilger quartz double monochromator and photomultiplier tube. This was particularly important at the longer wavelengths (546 and 578 $m\mu$; see below) where a small transmission of shorter wavelength radiation would give rise to large errors in the quantum efficiency. The stray light was in most cases appreciable, as is inevitable with the filters available, and its amount is indicated in table 1. Because of the stray light, photolyte transmission values, where these were appreciable, were measured using the actual filtered beam by means of a phototube situated behind the constant-temperature block. The observed extent of photo-decomposition was corrected for light reflected from the back surface of the cell, and for the ferrioxalate solutions an additional correction was applied to allow for the slight change in transmission caused by the decomposition.

For the determinations at 366, 334, 313 and 302 $m\mu$ the light from the 1 kW high-pressure mercury lamp (operated from a rectified three-phase supply) was focused by means of a 10 in. mirror on to the entrance slit of the D 96 monochromator. The entrance slit of the latter was opened to 3 mm, the exit slit to 1 mm. The diverging monochromatic beam passed to a quartz lens (5 cm focal length) and thence converged through a beam-splitter on to the fused silica actinometer cell (35 $\times$ 45 mm and 15 mm optical depth) contained in a constant-temperature block. The area of cell surface illuminated was 1 to 2 sq.cm. The proportion of stray light in the beam was appreciable, and the visible components were eliminated by the insertion of 2 mm of Chance OX 7 filter at the exit slit. To measure the purity of the resultant beam the actinometer cell was replaced by a silica right-angle prism which served to throw a horizontal image of the exit slit on to the entrance slit of the quartz double monochromator used to analyze the light. The proportion of stray light at each wavelength setting is shown in table 1. It is of interest to note that the 1 kW high-pressure lamp provides an intense emission at 334 $m\mu$ (approximately equal to that available at 313 $m\mu$) and provides a valuable source of ultra-violet light not normally available from medium-pressure lamps. The light from the high-pressure lamp was found to be less steady than that from the 125 W lamp and did not bear a direct relationship to the current passing. To obtain an accurate correction for this slow fluctuation the output from the phototube monitor was fed

TABLE 1. DETAILS OF LIGHT SOURCES USED

wavelength (m μ)	lamp used	method of isolation	spectral impurities	
			wavelength (m μ)	approximate %
577/9	125 W MB/D	filters A + C + D + E	< 560	0.01
			560-568	1.1
			> 588	1.4
546		filters A + F + G	< 525	0.05
			525-535	0.5
			> 557	6.7
			(mainly 577/9)	
436		filters A + H + J	< 410	1.6
			410-430	5.0
			440-460	4.7
			> 460	1.4
405		filters A + K + L	< 388	0.6
			388-402	4.5
			410-430	5.5
			> 430	3.5
			(mainly 436)	
365/6		filters A + B	< 340	0.4
			340-364	3.1
			370-394	4.9
			> 394	0.2
365/6	1 kW, mercury	D 96 monochromator	< 355	0.4
			355-363	5.3
			374-380	3.0
			> 380	0.5
334			< 324	0.6
			324-332	14.7
			341-348	7.3
			> 348	1.3
313			< 305	0.1
			305-311	3.5
			323-329	3.5
			> 329	0.9
297/302			< 292	0.4
			292-295	1.6
			305-312	13.9
			> 312	0.4
254	low-pressure mercury	filter M	265-280	0.6
			298-314	6.3
			> 314	< 0.1
see table 4	2½ kW Hg/Cd or 125 W MB/D	double monochro- mator	outside $\lambda \pm 5$	< 5

Filter solutions referred to in table 1: (A) 2 cm of 10% w/v $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ in 0.01N- H_2SO_4 in glass cell; (B) 2 mm of Chance glass OX1; (C) 1 cm of solution containing 10 ml. of $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ (1000 g in 1 l. of water) and 90 ml. of 3M- CaCl_2 ; (D) 2 cm of potassium dichromate solution (15 g in 200 ml. of water); (E) red gelatine filter (transmission 0% at 560 m μ , 50% at 580 m μ); (F) 1 cm of solution containing 50 ml. $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ (1000 g in 1 l. of water) and 50 ml. of 3M- CaCl_2 ; (G) 2 cm of a solution prepared from 50 ml. of solution D diluted to 250 ml. with water; (H) 2 cm of solution containing $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ 25 g and 300 ml. of ammonium hydroxide ($d=0.88$) made up to 1 l. with water; (J) 1 cm of solution containing 75 g NaNO_2 in 100 ml. of water; (K) 2 cm of solution containing 200 g $\text{Cu}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in 100 ml. of water; (L) 1 cm of solution containing 0.75 g of iodine in 100 ml. of carbon tetrachloride; (M) 4 cm of solution containing 321 g $\text{NiSO}_4 \cdot 7\text{H}_2\text{O}$ and 56 g of $\text{CoSO}_4 \cdot 7\text{H}_2\text{O}$ /l.

TABLE 2. QUANTUM EFFICIENCY BY REFERENCE TO URANYL OXALATE

Temp. 22°C; 15 ml. of 0.006M- $K_2Fe(C_2O_4)_3$ in 0.1N-sulphuric acid; optical depth 15 mm; solutions deaerated and stirred by means of a current of oxygen-free nitrogen; fraction of light absorbed was 61.5% at 436 m μ , 96.2% at 405 m μ and 100% at other wavelengths.

wave-length (m μ)	light source	time of exposure		mean rate of decom- position ferrioxalate (calcd. for 100% ab- sorption) 10 ⁻⁶ mole/ (min)	quantum efficiency		mean quantum efficiency, ferrioxalate
		uranyl oxalate (min)	ferri- oxalate (min)		assumed for uranyl oxalate	observed for ferri- oxalate	
436	MB/D with filters	300	1.5	2.049	0.58	1.121	1.11(6)
		300	1.0	2.069		1.102	
		300	1.0	2.085		1.126	
		300	1.0	2.068		1.114	
405	MB/D with filters	300	2.0	0.5349	0.56	1.188	1.13(9)
		300	2.0	0.5431		1.053	
		480	2.0	0.5433		1.142	
		420	2.0	0.5518		1.171	
365/6	MB/D with filters (Parker 1953) (see part I)	360	1.0	2.20	0.49	1.18	1.16(0)
		360	1.0	2.22		1.17	
		360	1.0	2.05		1.18	
		360	1.0	2.03		1.18	
	1 kW with D 96	180	4.0	3.361		1.128	
		210	4.0	3.264		1.143	
		210	4.0	3.258		1.138	
334	1 kW with D 96	270	4.0	1.230	0.53	1.256	1.25(2)
		270	5.0	1.173		1.207	
		270	5.0	1.230		1.246	
		240	2.0	1.215		1.282	
		270	3.0	1.912		1.272	
		270	3.0	1.776		1.242	
		240	3.0	1.529		1.257	
		270	3.0	1.552		1.256	
313	1 kW with D 96	300	4	2.064	0.56	1.265	1.22(9)
		300	4	1.869		1.201	
		240	4	2.040		1.209	
		240	4	1.924		1.215	
		270	1.5	1.743		1.235	
		240	6	1.984		1.236	
		240	3	1.884		1.226	
		240	3	1.776		1.239	
297/302	1 kW with D 96	210	3	1.752		1.235	1.23(8)
		270	4	1.609	0.57	1.238	
		270	4	1.498		1.223	
		270	4	1.597		1.273	
		270	4	1.538		1.231	
254	low-press. Hg (see part I)	270	4	1.507		1.226	1.22(3)
		300	2	1.437	0.60	1.24	
	low-press. Hg with filter	150	2	1.526		1.23	
		360	10	0.1414		1.232	
		300	10	0.1379		1.191	

after amplification to a recorder and to a voltage integrating motor. The readings from both the chart and the motor were checked periodically against a microammeter in the phototube circuit.

The uranyl oxalate solution contained 0.01 M-uranyl oxalate in 0.05 M-oxalic acid. The oxalate consumption was determined by weight titration with 0.04 N-potassium permanganate solution standardized daily against weighed portions of pure sodium oxalate. Portions of unexposed uranyl oxalate solution were titrated daily and it was found to maintain a constant concentration for long periods. Solutions were titrated at 60 to 70°C with the sulphuric acid concentration held at about 0.6 N. Portions of ferrioxalate solution were exposed at the beginning and end of the uranyl oxalate irradiation, and the amount of ferrous iron formed was determined by the method previously described (Parker 1953). The mean rates of decomposition of the two solutions, after correction for fluctuation in light intensity, transmitted light, etc., were used to obtain the quantum efficiency of the ferrioxalate solutions by using the values for uranyl oxalate reported by Leighton & Forbes (1930). The values of quantum efficiency obtained with 0.006 M-potassium ferrioxalate in 0.1 N-sulphuric acid are shown in table 2. All solutions were stirred during irradiation by the passage of oxygen-free nitrogen.

QUANTUM EFFICIENCY BY REFERENCE TO THERMOPILE

Independent values of quantum efficiency at 436, 405 and 366 m μ were obtained by direct reference to a Cambridge 18-junction thermopile calibrated by the National Physical Laboratory. The effective area of the thermopile surface was approximately 0.4 sq.cm. The thermopile output was measured by means of a Doran potentiometer via a voltage divider which allowed the potentiometer to measure potentials in the thermopile-galvanometer circuit to within $\pm 0.05 \mu\text{V}$. A Cambridge short-period galvanometer was used and the sensitivity of the circuit was 8 mm/ μV at 1 m. The body of the thermopile was lagged with cotton wool and the drift was small. Zero readings were always taken, with the light shutter closed, both before and after each measurement, and the mean deflexions of six such measurements were used to calculate the light intensity.

Owing to the great sensitivity of the ferrioxalate actinometer, lower light intensities could be used in these experiments because the limit was set by the sensitivity of the thermopile-galvanometer circuit instead of by that of the uranyl oxalate. Advantage was taken of this in the design of the optical arrangements, in which intensity was sacrificed to obtain a more uniformly collimated beam from the 125 W mercury lamp. After collimation the beam passed through the 2 cm layer of 10 % w/v $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ solution and the double-compartment cell containing the appropriate filter solution. The phototube monitoring device was again used.

A uniform area in the centre of the beam was chosen for the experiment. Movement of the thermopile in any direction from the position by 0.5 cm produced less than 2 % variation in the reading. It was arranged that the thermopile could be readily replaced by the actinometer cell in its constant-temperature block, a portion of the beam being selected by means of a thin metal mask having an accurately

machined aperture 1 cm in diameter. The centre of the aperture could be arranged to coincide exactly with the centre of the sensitive area of the thermopile, and the mean thermopile readings obtained in one series of experiments could be used (after correction for fluctuation of light intensity, as indicated by the phototube monitor unit, and for absorption and reflexion by the front surface of the cell as determined by transmission measurements when the cell was filled with water) to calculate the intensity falling on the aperture during the irradiation of the ferrioxalate. In

TABLE 3. QUANTUM EFFICIENCY BY REFERENCE TO N.P.L.—
CALIBRATED THERMOPILE

Conditions as in table 2; light source was MB/D lamp with appropriate filters (see table 1); area of cell face exposed was 0.785 sq.cm.; fractions of light transmitted by front surface of cell were respectively 0.957, 0.951 and 0.933.

wavelength (m μ)	mean thermopile response (μ V)	intensity at aperture (μ W/cm ²)	time of exposure of ferrioxalate solution (min)	rate of reaction (calcd. for 100 % absorption and 100 % cell face transmission) (10 ⁻⁶ mole/min cm ²)	quantum efficiency
436	22.69	403.4	30	0.1013	1.148
	24.93	443.2		0.1123	1.158
	25.81	458.9		0.1145	1.141
	19.24	342.0		0.07878	1.053
	18.77	333.7		0.07838	1.074
	18.86	335.3		0.07878	1.074
	19.65	349.3		0.08380	1.097
	19.70	350.2		0.08500	1.110
					mean 1.10 (7)
405	5.37	95.5	45	0.02241	1.155
	5.10	90.7	30	0.02152	1.168
	5.34	94.9	45	0.02216	1.149
	5.08	90.3	45	0.02119	1.155
					mean 1.15 (7)
365/6	14.28	253.9	15	0.05973	1.281
	14.34	254.9		0.06004	1.283
	14.46	257.1		0.05850	1.239
	14.46	257.1		0.05880	1.246
	11.42	203.0	20	0.04628	1.242
	11.53	205.0		0.04761	1.265
	11.29	200.7		0.04684	1.271
					mean 1.26 (1)

this way the necessity for integration of the total beam by traversing the thermopile was avoided (this is probably one of the major sources of error in previous calibration of the uranyl oxalate system). This method has the further advantage that variations in sensitivity over the thermopile surface introduce no errors. For each determination of quantum efficiency a series of six thermopile readings against phototube monitor readings was taken both before and after the exposure of the ferrioxalate solution. The results obtained at 366, 405 and 436 m μ are shown in table 3.

The intensity available in the collimated beam used for the thermopile experiments at 436, 405 and 366 $m\mu$ was not sufficient for the experiments at 546 and 578 $m\mu$ where the quantum efficiency and light absorption are both low. The focused beam arrangement already described was therefore employed. The filters used for isolation of the 546 and 577/579 $m\mu$ mercury lines are indicated in table 1. The spectral purity was checked by means of the double monochromator with photo-multiplier attachment. The thermopile was placed at a distance of 20 cm from the

TABLE 4. COMPARISON OF QUANTUM YIELDS AT DIFFERENT WAVELENGTHS USING THERMOPILE

Temp. 22° C; value at 436 $m\mu$ assumed to be 1.11; concentration of potassium ferrioxalate was 0.15 M for wavelengths ≥ 469 and 0.006 M for wavelengths $\leq 405 m\mu$.

wave-length ($m\mu$)	source and method of isolation	approx. intensity (10^{-6} E $\text{min}^{-1} \text{cm}^{-2}$)	fraction of light absorbed	observed relative quantum yields	mean quantum yield
577/9	MB/D, filters	0.14	0.118	0.013, 0.013	0.013
546		0.29	0.061	0.147, 0.152, 0.157	0.15 (2)
509	Hg/Cd, d.m.	0.022	0.132	0.848, 0.868, 0.863	0.86 (0)
480		0.013	0.578	0.947, 0.939, 0.929	0.93 (8)
468		0.008	0.850	0.933, 0.921, 0.920	0.92 (5)
436 (0.15 M)	MB/D, filters	0.62	0.997	1.033, 1.036, 1.015	} 1.01 (2)
	Hg/Cd, d.m.	0.007		1.015, 1.000, 0.990, 0.995	
436 (0.006 M)	MB/D, filters	0.62	0.615	—	} [1.110]
	Hg/Cd, d.m.	0.007	0.610		
405	MB/D, filters	0.16	0.962	1.130, 1.166, 1.175, 1.164, 1.148, 1.150	} 1.13 (3)
	Hg/Cd, d.m.	0.003	0.976	1.103, 1.117, 1.108, 1.125, 1.110, 1.105	
365/6	MB/D, filter	0.21	> 0.999	1.220, 1.199, 1.209, 1.210	} 1.21 (6)
361/6	Hg/Cd, d.m.	0.010		1.219, 1.224, 1.225, 1.224	
334	MB/D, d.m.	0.001		1.233, 1.181, 1.228, 1.193	1.20 (9)
313	MB/D, d.m.	0.006		1.264, 1.252, 1.247	} 1.24 (4)
	Hg/Cd, d.m.	0.003		1.210, 1.247	
254	MB/D, d.m.	0.001		1.294, 1.236, 1.290, 1.328	1.28 (7)

actinometer aperture, so that, being comparatively distant from the focus of the beam, its reading was proportional to the total energy passing through the aperture irrespective of the filters employed. Experiments were made at 405 and 436 $m\mu$ as well as at 546 and 578 $m\mu$. The rate of decomposition at 436 $m\mu$ was used to calibrate the thermopile in terms of total energy flux through the aperture, and quantum efficiencies at the other wavelengths were thus calculated. Where necessary corrections for transmitted light, etc., were applied as previously described. The results are recorded in table 4.

The determination of quantum efficiencies at additional wavelengths in the visible region was made possible by the acquisition of a 2½ kW mercury-cadmium lamp which was operated from rectified three-phase supply. The light was passed

through 2 cm of water and then focused on the entrance slit of a quartz double monochromator. The monochromatic light emerging from the exit slit passed through the beam-splitter of the monitor and thence to a concave mirror which served to focus the whole of the light beam on to the sensitive surface of the 18-junction thermopile. The actinometer cell (15 mm optical depth) in its constant-temperature block could be interposed in the beam between the mirror and the thermopile (the illuminated area of the front surface of the actinometer cell was approximately 2 sq.cm). Where necessary the transmission of the photolyte was measured by means of a phototube placed behind the actinometer cell. A series of ten thermopile deflexions (zero readings were taken by shuttering the light beam at the intermediate slit) were measured simultaneously with phototube monitor readings immediately before and again immediately after the irradiation. The overall mean values obtained served to convert the mean monitor reading during the irradiation to a relative energy. On the same day a similar set of readings was made with the wavelength set at 436 m μ , and at this wavelength setting an irradiation using 0.006 M-potassium ferrioxalate in 0.1 N-sulphuric acid served to calibrate all thermopile readings at the other wavelengths in terms of absolute intensity values. This calculation was justifiable because, from the entrance slit of the double monochromator to the actual thermopile surface, no lenses were used for focusing. Slight corrections were, however, applied for the changes in transmission values with wavelength of the thermopile window and the front surface of the cell. The results of runs carried through on different days are shown in table 4.

Relative values of quantum efficiency were also obtained in the same manner at 364 and 313 m μ using the 2½ kW mercury-cadmium lamp and at 334 and 254 m μ using the 125 W mercury lamp (type MB/D without glass envelope).

TEMPERATURE COEFFICIENTS

During the course of the above determinations, comparisons were made of the rates of photolysis of ferrioxalate solutions held at different temperatures in the constant-temperature block. Care was taken to measure the photolyte transmission at the appropriate temperature, and in some cases appreciable changes with temperature were observed, presumably caused by changes in the equilibrium concentrations of the various photo-active oxalate complexes present (Parker 1954). The resulting temperature coefficients are recorded in table 6. At all wavelengths below 509 m μ the coefficients per 10° C are small, and for most purposes accurate temperature control when using the actinometer is unnecessary provided that depth and concentration of actinometer liquid are chosen so that light absorption is high.

SUMMARY OF QUANTUM EFFICIENCIES

The mean quantum yields for 0.006 M-potassium ferrioxalate in 0.1 N-sulphuric acid determined by reference to uranyl oxalate are compared in table 5 with the mean values obtained by reference to the thermopile. In calculating the values in table 4 and in column 4 of table 5 a value for the quantum efficiency at 436 m μ had to be

chosen for reference. The value of 1.11, obtained by absolute measurement with the thermopile, was chosen. On this basis the results obtained by the different methods are in very good agreement, except at 366 $m\mu$ and, to a less extent, at 254 $m\mu$. In comparing the results it should be remembered that the quantum efficiencies reported by Leighton & Forbes (1930) for 0.01 M-uranyl sulphate in 0.05 M-oxalic acid have been used (at 334 $m\mu$ an interpolated value had to be used). The accuracy of the quantum yields shown in column 2 of table 5 therefore depends entirely on

TABLE 5. 0.006 M-FERRIOXALATE: SUMMARY OF MEAN QUANTUM YIELDS OBTAINED BY DIFFERENT METHODS

wavelength ($m\mu$)	by comparison with uranyl oxalate	by reference to calibrated thermopile	by comparison with ferri- oxalate at 436 $m\mu$ using thermopile	recommended value
436	1.12	1.11	[1.11]	1.11
405	1.14	1.16	1.13	1.14
{ 365/6 361/6	1.16 —	1.26 —	1.22	1.21
334	1.25	—	1.21	1.23
313	1.23	—	1.24	1.24
297/302	1.24	—	—	1.24
254	1.22	—	1.29	1.25

TABLE 6. TEMPERATURE COEFFICIENTS BETWEEN 22 AND 32°C

wavelength ($m\mu$)	concentration of ferrioxalate (M)	rate at 32°C rate at 22°C (corrected for light absorption)	mean temp. coefficient (%/10°C)
509	0.15	1.168, 1.163, 1.138	+ 16
480	0.15	1.031, 1.043	+ 4
468	0.15	0.987, 1.016	nil
436	0.15	1.012, 0.998, 1.006, 0.992	nil
436	0.006	0.986, 0.998, 1.027, 1.018	+ 1
405	0.006	1.048, 1.042, 1.048	+ 5
361/6	0.006	1.000, 1.033, 1.032, 0.979, 1.027, 1.011	+ 1
334	0.006	0.998, 0.981, 0.987, 0.965	- 2
313	0.006	1.009, 1.022, 1.027, 1.014	+ 2
297/302	0.006	1.009, 1.025, 1.022, 1.016	+ 2

the accuracy of Leighton & Forbes's mean values. The results of their individual determinations differed quite appreciably. Thus, at 254 $m\mu$ two values were reported (0.58 and 0.63), while at 366 $m\mu$ fourteen results were reported between 0.445 and 0.53. Independent determinations at 254 $m\mu$ were made by Brackett & Forbes (1933) who reported a mean result of 0.62. Had this value been used for calculating the result shown in table 5 satisfactory agreement with the thermopile figure would have been obtained. It should also be mentioned that comparison between the two actinometers by Baxendale & Bridge (1955) gave a mean value at 366 $m\mu$ of 1.20 for the ferrioxalate actinometer, in closer agreement with the

thermopile results shown in table 5. Their results at other wavelengths were also in reasonable agreement (1.07 at 436 m μ and 1.25 at 254 m μ).

The recommended values shown in column 5 of table 5 are considered to be within a small percentage of the true values. Better accuracy than this could only be obtained by an exhaustive investigation using a number of independently calibrated thermopiles. Mean values of quantum efficiency at wavelengths longer than 436 m μ (for which uranyl oxalate comparisons are not available) are given in table 4. These refer to 0.15 M solution because the light absorption by the weaker solution is inconveniently low at long wavelengths.

EFFECT OF LIGHT INTENSITY ON THE QUANTUM EFFICIENCY AT 366 m μ

Light from a 250 W compact source high-pressure mercury lamp was concentrated by lenses on to a circular aperture 1 cm in diameter. The out-of-focus image of the source was somewhat larger than the aperture, and the intensity of light across the latter was approximately uniform. Immediately behind the aperture was placed the actinometer cell in its constant-temperature block. The 366 m μ group of mercury lines was isolated by means of 2 cm of 10 % w/v CuSO₄.5H₂O solution and 2 mm of Chance glass OX 1. Owing to the diffuse nature of the spectral lines and the high background intensity from the lamp, the filtered light was much more polychromatic than that from the 125 W lamp previously used. Of the total energy 80 % was accounted for in the region 362 to 377 m μ and a high proportion (13 %) was in the region 377 to 391 m μ . However, the beam was considered to be sufficiently monochromatic for the investigation of the effect of variation of intensity. A negligible proportion of the beam was transmitted by the 15 mm layer of photolyte. The light intensity was monitored in the usual manner.

The intensity of light falling on the aperture could be reduced by known amounts by the insertion of neutral filters in the beam. The effective transmission of the filters used was measured by means of a phototube fitted with a neutral filter, which was situated on the far side of the aperture and received the whole of the beam passed by the aperture via a lens. Four filters in all were used (see table 7), two consisting of single sheets of fine copper gauze and having transmission values of about 0.34, and two consisting of pairs of sheets, soldered to a copper spacer with their meshes at an angle of 45°. The latter had transmission values of the order 0.1.

The results are shown in table 7. In experiments 2 to 5 the position of the aperture was kept constant and the light intensity was reduced by the insertion of the filters. The observed rates of decomposition are shown in column 4, and the relative light intensities calculated from the measured transmission values of the filters are given in column 5. Within experimental error the ratio of the decomposition rate to the light intensity was constant, showing that the quantum efficiency was independent of intensity over this range. The filters were then removed and the aperture and actinometer cell moved to a greater distance, so that the intensity was reduced to a value somewhat greater than that measured in experiment 5. The actual relative intensity was then calculated from the observed rate of decomposition. The intensity was then further reduced by the filters. Again, the ratio of decomposition rate

to intensity was constant within experimental error. It was therefore concluded that over a range of intensities from 0.003 to $6.5 \mu\text{E}/\text{cm}^2 \text{ min}$, the quantum efficiency remained constant.

That the quantum efficiency is independent of light intensity was demonstrated by an entirely different method which did not depend upon the precise measurement of filter transmission values. The optical arrangement was similar to that used in the previous experiments, but with the following modifications. The filtered light from the compact source lamp was brought to a focus just short of the 1 cm diameter aperture. Thus the light passing through the aperture diverged in a reasonably uniform manner. Exposures were made using a large glass cell 11 cm in diameter. First, several exposures were made with the cell against the aperture, but

TABLE 7. EFFECT OF VARYING THE LIGHT INTENSITY
USING COPPER GAUZE SCREENS

screens used	transmission of screens	time of exposure (min)	rate of decomposition (10^{-6} mole/min)	relative light intensity	rate of decompn light intensity
none	—	2	6.18)	1.000	6.20
		0.5	6.22)		
no. 1 (single)	0.3432	0.5	2.128	0.3432	6.20
no. 2 (single)	0.3426	2	2.101	0.3426	6.13
no. 3 (double)	0.0903	2	0.560	0.0903	6.20
no. 4 (2 double)	0.01045	10	0.0664	0.01045	6.35
none	—	5	0.243	0.0392	—
no. 3 (double)	0.0903	30	0.0223	0.00354	6.30
no. 4 (2 double)	0.01062	120	0.00267	0.000416	6.42

with different areas of the cell surface illuminated. In this way it was shown that the transmission by the cell face was uniform. Exposures were then made with the cell withdrawn so that the area illuminated covered a circle approximately 7 cm in diameter. Thus in both sets of experiments the whole of the beam was intercepted by the actinometer liquid, but in the latter set the light intensity was reduced by about 50 times. The rate of decomposition was unaltered (2.59 , 2.58 and 2.57×10^{-6} mole/min with cell against the aperture and 2.59 and 2.58×10^{-6} mole/min with the cell withdrawn).

To determine the quantum efficiency at higher intensities than that available from the compact source lamp a spiral quartz xenon-filled flash discharge tube was used. This was operated by a bank of condensers ($328 \mu\text{F}$) charged to 2000 V. The time for 50 % discharge was about $750 \mu\text{s}$. The procedure was similar to that used by Christie & Porter (1952), except that a composite gauze filter of blackened copper was used for reducing the light intensity. Experiments were made both with the full light from the flash tube and also with ultra-violet light selected by means of 2 mm of Chance glass OX 1. In each case 15 ml. of 0.006 M-potassium ferrioxalate in 0.1 N-sulphuric acid was exposed (after de-oxygenation) in a rectangular silica cell of 15 mm optical depth situated immediately behind a mask having an aperture of area either 8.3 or 0.79 sq.cm. The cell was screened from all light except that

passing through the aperture. Exposures were made at three distances from the source both with and without the neutral filter in position. In each experiment the number of flashes was adjusted to provide an adequate quantity of ferrous iron for spectrophotometric measurement. The number of flashes varied from 1 in the nearest position without filters to 150 in the most remote position with both the neutral and OX 1 filters in position.

TABLE 8. VARIATION OF INTENSITY USING XENON FLASH DISCHARGE TUBE

The transmission of the neutral gauze filter, measured photoelectrically, was 0.071. In column 4 the duration of flash is assumed to be 1 ms and the mean quantum yield 1.2. No account has been taken of light not absorbed.

distance from lamp (cm)	area of cell face exposed (cm ²)	wavelength filter used	approximate 'actinic' inten- sity on surface of liquid (without neutral filter) (10 ⁻⁶ E min ⁻¹ cm ⁻²)	decomposition per flash (10 ⁻⁶ mole)		apparent trans- mission of neutral filter
				without neutral filter	with neutral filter	
76	8.3	OX 1	58	0.00963	0.000686	0.071
76	8.3	none	250	0.0410	0.00293	0.071
25	8.3	OX 1	480	0.0810	0.00602	0.074
				0.0784	0.00563	0.072
25	8.3	none	1900	0.318	0.0223	0.070
				0.316	0.0230	0.073
8	8.3	OX 1	3200	0.511	0.0362	0.071
				0.545	0.0354	0.065
8	0.79	OX 1	3100	0.0485	0.00381	0.079
				0.0481	0.00347	0.072
8	8.3	none	12500	2.09	0.148	0.071
				2.06	0.146	0.071
8	0.79	none	12400	0.203	0.0141	0.069
				0.190	0.0145	0.076

The intensity at the surface of the liquid under the latter conditions corresponded to approximately 4 $\mu\text{E}/\text{cm}^2 \text{ min}$, i.e. somewhat less than the highest intensity investigated with a continuously burning lamp. Removal of the neutral filter increased this intensity by about 14 times. To determine whether the quantum efficiency had changed at this higher intensity the exact ratio of intensities (as determined from the known transmission value of the filter) was compared with the observed ratio of the amounts of ferrous iron formed. The procedure was then repeated at the higher intensities in the position nearer to the lamp. The results are shown in table 8. The reproducibility of the decomposition ratios in column 7 is not high (probably owing to fluctuation in the output from the flash tube), but the results are sufficiently close to the actual transmission value of the filter to indicate that there is little change in quantum efficiency up to an intensity equivalent to about 12 000 $\mu\text{E}/\text{min}$. Separate experiments at the highest intensity (OX 1 filter) demonstrated that oxygenation produces a negligible effect.

VARIATION OF PHOTOLYTE COMPOSITION

Previous work (Parker 1954) had shown that in 0.1 N-sulphuric acid containing potassium ferrioxalate, at least two oxalato-ferric complexes are present, and that by the addition of further ferric iron a solution could be produced which contained substantially only the one complex.

The quantum efficiency of the latter at 366 m μ was found to be equal to that of the 'normal' actinometer solution containing the mixture of complexes. In the absence of oxygen, the quantum efficiency in neutral solution was also found to have the same value. Further variations of the photolyte composition have now been investigated, particularly from the point of view of the sensitivity of the quantum efficiency to photolyte composition when the system is used as an actinometer. All irradiations were made in cells of 15 mm optical depth containing 15 ml. of

TABLE 9. QUANTUM EFFICIENCY AT 366 m μ AS A FUNCTION OF SOLUTION COMPOSITION

K ₃ Fe(C ₂ O ₄) ₃ (M)	acid	added salt	rates of reaction (10 ⁻⁶ mole/min)	quantum efficiency
0.006	0.1 N-H ₂ SO ₄	—	1.88, 1.86	(1.21)
0.0006		—	1.81, 1.80	1.17
0.06		—	1.81, 1.83	1.18
0.15		—	1.77, 1.78	1.15
0.006	10 N-H ₂ SO ₄	—	0.24, 0.14	0.12
	1.0 N-H ₂ SO ₄	—	1.77, 1.75	1.14
	0.01 N-H ₂ SO ₄	—	1.82, 1.84	1.18
0.006	0.1 N-HCl	—	1.85	1.20
	0.1 N-HClO ₄	—	1.87	1.21
	0.1 N-H ₂ SO ₄	1.0 M-Na ₂ SO ₄	1.73, 1.74	1.12
	0.1 N-HCl	1.0 M-NaCl	1.74, 1.77	1.14
	0.1 N-HClO ₄	1.0 M-NaClO ₄	1.71	1.11

photolyte. Light from a 125 W lamp, type MB/D, was filtered by 2 cm of 10 % w/v CuSO₄·5H₂O and 2 mm of Chance glass OX 1. The filtered light was focused on a 2 × 2 cm aperture placed immediately in front of the actinometer cell in its constant temperature block. The temperature was 22°C throughout, and oxygen-free nitrogen was passed before and during photolysis.

The effect of variation of ferrioxalate concentration is shown in table 9 (experiments 1 to 4). At 0.006 M and above light absorption is complete. The observed rate of decomposition in the most dilute solution was corrected for the fraction of light transmitted as measured by a phototube situated behind the actinometer cell. Allowance was made for reflexion from the rear surface of the cell and for the small change in absorption during irradiation. Table 9 shows that variation in concentration by a factor of 250 has little effect on the quantum efficiency. The value in the strongest solution is reduced slightly.

The effect of varying sulphuric acid concentration is shown in table 9 (experiments 5 to 7). Between 1.0 and 0.01 N the quantum yield varies only slightly. It is a maximum in the region of 0.1 N, i.e. the concentration chosen for the standard

actinometer. At very high acidities the yield falls considerably, due to dissociation (with production of ferric iron and its sulphate complexes which act as inner filters). Table 9 also shows the results obtained using hydrochloric and perchloric acids in place of sulphuric acid (experiments 8 and 9) and the effect of adding a large excess of neutral electrolyte (experiments 10 to 12). The quantum yield is independent of the anion present, but is reduced slightly at high ionic strength.

Some less exhaustive experiments at 313 $m\mu$ have shown that, at this wavelength also, the quantum yield of ferrous iron is not critically dependent on the concentrations of ferrioxalate or sulphuric acid in the region of 0.006 M and 0.1 N respectively.

EFFECT OF ADDED FERROUS IRON

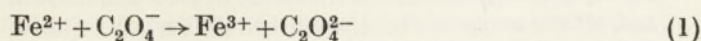
Evans, Santappa & Uri (1951) have shown that during the photolysis of certain ferric iron complexes both primary and secondary dark back reaction can occur after the initial photo-excitation. The incidence of the secondary dark back reaction results in a reduction in quantum yield when ferrous iron is added to the solution. If a secondary dark back reaction occurred in the case of the ferrioxalate actino-

TABLE 10. EFFECT OF ADDED FERROUS IRON ON THE RATE OF PHOTOLYSIS AT 366 $m\mu$

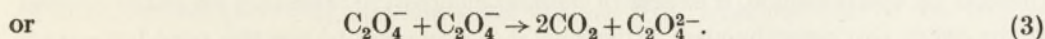
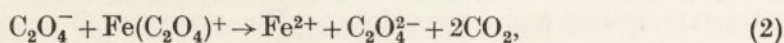
In each experiment 15 ml. of 0.006 M ferrioxalate in 0.1 N- H_2SO_4 was photolyzed for 15 min while stirred by a current of oxygen-free nitrogen.

Fe ²⁺ added (10 ⁻⁶ mole)	Fe ²⁺ formed by photolysis (10 ⁻⁶ mole)	mean concentration of Fe ²⁺ during photolysis (10 ⁻³ M)
nil	39.2, 38.4	1.29
15	39.4, 38.9	2.30
30	38.5, 38.2	3.28
60	39.1, 40.1	5.32
120	39.3, 39.5	9.31

meter the accumulating ferrous iron during prolonged irradiation would cause a steady decrease in quantum yield which would be most objectionable. In preliminary experiments (Parker 1953) no sign of this effect was observed. Further experiments have since been made in which ferrous iron was added to the solution before photolysis, and the results (see table 10) indicate that secondary dark back reaction is negligible up to concentrations of ferrous iron somewhat greater than the concentration of ferrioxalate initially present. It would appear therefore, that, within this concentration range, the reaction



does not occur to an appreciable extent. The intermediate oxalate radical ($C_2O_4^{\cdot -}$ or $CO_2^{\cdot -}$) must therefore be consumed by other processes, e.g.:



Reaction (3) would be favoured by high light intensities, and its incidence would result in a reduced quantum yield of ferrous iron. No indication of this was obtained in the present experiments at high intensity (table 8), although at the highest intensity with the OX 1 filter the rate of production of intermediate radicals must have been of the order of 3×10^{-4} mole/l. ms in the 1 mm depth of liquid in which the light was mainly absorbed. However, the present experiments with the flash tube were not sufficiently precise to detect small changes in quantum yield, and the investigation is being continued by the method of flash photolysis. A detailed discussion of the proposed mechanism of the reaction and of the effect of oxygen in neutral solution has been given elsewhere (Parker 1954).

RECOMMENDED PROCEDURES FOR USING THE FERRIOXALATE ACTINOMETER

Preparation of actinometer liquid and choice of concentration and optical depth

Pure potassium ferrioxalate can be readily prepared by mixing 3 volumes of 1.5 M A.R. potassium oxalate with 1 volume of 1.5 M A.R. ferric chloride with vigorous stirring. The precipitated potassium ferrioxalate should be recrystallized, preferably three times, from warm water and the crystals sucked dry and dried in a current of air at 45°C. The composition corresponds to $K_3Fe(C_2O_4)_3 \cdot 3H_2O$. The 0.006 M actinometer solution is prepared by dissolving 2.947 g of the crystals in 800 ml. of water. 100 ml. of 1.0 N-sulphuric acid are added, the solution diluted to 1 l. and mixed. The 0.15 M solution is made in a similar manner using 73.68 g of ferrioxalate crystals.

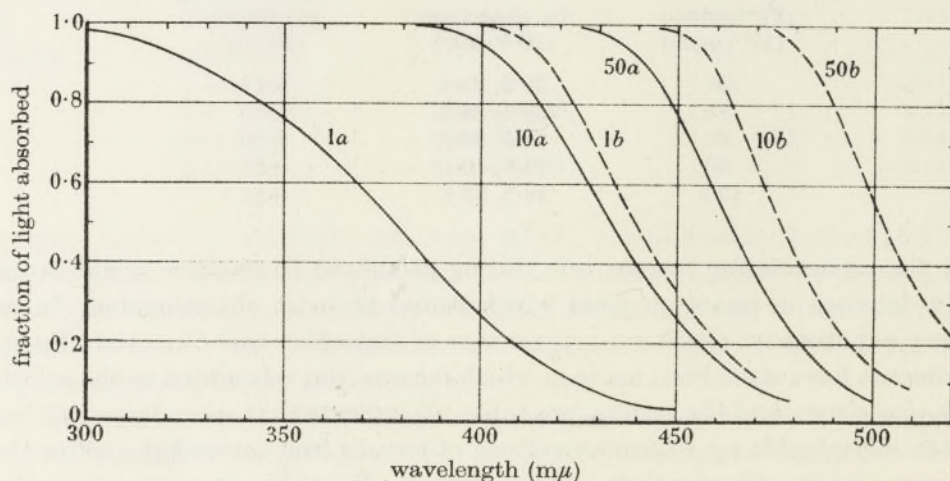


FIGURE 1. Light absorption by recommended actinometer solutions. Curves 1a, 10a and 50a are for 0.006 M-potassium ferrioxalate in 0.1 N-sulphuric acid with optical depths of 1, 10 and 50 mm respectively. Curves 1b, 10b and 50b are for 0.15 M-potassium ferrioxalate in 0.1 N-sulphuric acid at the same optical depths respectively.

The actinometer solutions could be made up by mixing standard solutions of potassium oxalate and ferric sulphate. The quantum efficiency at 366 mμ is little affected by the presence of excess of neutral electrolyte, and this probably applies at the other wavelengths also. However, it is preferred to use the pure potassium

ferrioxalate, because this forms a very convenient standard substance. It is readily prepared in pure state, it has an accurately known composition, and it can be stored indefinitely without change.

A 1 cm depth of 0.006 M solution absorbs 99 % or more of the light of wavelengths up to 390 m μ . It can be used conveniently for wavelengths up to about 430 m μ (absorption about 50 %/cm). For longer wavelengths the 0.15 M solution is usually more convenient. The fractions of light absorbed by various thicknesses of the two solutions are shown in figure 1 and can be used as a guide to indicate the required depth and concentration for any particular purpose. When very accurate results are required and light absorption is not complete it is desirable to measure directly the fraction of light absorbed by the actual layer of liquid used, preferably using the same source of light, because the fraction absorbed will in general vary with temperature, and, of course, with the spectral purity of the light.

When stored in an amber bottle in the dark room the actinometer solutions can be kept unchanged for long periods. The 'blank' values obtained by the method described below correspond respectively to optical densities of approximately 0.01 and 0.025/cm for 10 ml. after reaction and dilution to 20 ml.

Preparation of calibration graph for ferrous iron

Solutions required are: (a) 0.4×10^{-6} mole/ml. of Fe^{2+} in 0.1 N- H_2SO_4 (freshly prepared by dilution from standardized 0.1 M- FeSO_4 in 0.1 N- H_2SO_4), (b) 0.1 % 1 : 10-phenanthroline monohydrate in water, (c) buffer solution (600 ml. of N-sodium acetate and 360 ml. of N- H_2SO_4 diluted to 1 l.).

Into a series of 20 ml. calibrated flasks add the following volumes of solution (a): 0, 0.5, 1.0, ..., 4.5, 5.0 ml. Add 0.1 N- H_2SO_4 so as to make the total acidity equivalent to 10 ml. of 0.1 N- H_2SO_4 . Add 2 ml. of solution (b), 5 ml. of solution (c) (mixing after each addition), make up to mark, mix and allow to stand for at least $\frac{1}{2}$ h. Measure the optical densities at 510 m μ in a 1 cm cell with a spectrophotometer. Correct each optical density for the value obtained with the solution to which no ferrous iron had been added and plot the results against the amount of ferrous iron added.

Using a spectrophotometer of type similar to the Unicam 500 or 600 it has been found that 1.81×10^{-6} mole of ferrous iron in the 20 ml. test solution produces an optical density of 1.0/cm at 510 m μ . The optical density of the 'blank' solution was less than 0.01/cm, and a linear calibration graph was obtained. Using either of these spectrophotometers the calibration was found to remain constant during several years. In view of this and the fact that the graph is linear, it is possible to use a spectrophotometric factor to convert optical density to amounts of ferrous iron and only occasional checks of the calibration graph are required. With a filter photometer somewhat different values of 'blank' and graph slope may be obtained. Whatever the instrument, the user must, of course, prepare his own calibration graph to allow for instrumental variation. The 'difference optical density' at 510 m μ is independent of the presence of potassium ferrioxalate and of the volume of buffer added, provided that the latter is equal to at least one-half of the volume of 0.1 N-sulphuric acid present in the solution (final pH = 3.5). Half an hour is

more than sufficient to allow for complete reaction after addition of the phenanthroline. If desired the solutions may be kept for several hours (in the dark) before measurement.

General procedure for actinometry

Irradiate the appropriate actinometer solution for a period sufficient to produce a concentration of ferrous iron between 0.005×10^{-6} and 3×10^{-6} mole/ml. (During irradiation it is desirable to stir the liquid by means of a current of oxygen-free nitrogen.) After irradiation and mixing, pipette an aliquot of the solution into a 20 ml. calibrated flask. Add in succession 2 ml. of phenanthroline solution (*b*) and a volume of buffer (*c*) equal to half the volume of photolyte taken (mixing after each addition). Make up to the mark with water, mix and allow to stand for at least $\frac{1}{2}$ h. Measure the optical density at 510 m μ in either a 1 or 4 cm cell. Repeat with the same volume of unexposed actinometer solution. Convert the 'difference optical density' to quantity of ferrous iron using the calibration graph or spectrophotometric factor as appropriate. Convert the quantity of ferrous iron formed in the total volume of irradiated solution to a radiation dose using the recommended quantum efficiency recorded in table 4 (436 to 510 m μ) or table 5 (254 to 436 m μ). If necessary allow for the fraction of light absorbed (figure 1) to obtain the dose incident on the surface of the liquid. For concentrations up to 0.02×10^{-6} mole/ml. a 10 ml. aliquot should be taken and measurement made in a 4 cm cell. For higher concentrations an aliquot should be taken such that the optical density in a 1 cm cell is between 0.1 and 0.8. The amount of ferrioxalate decomposed should correspond to a reduction in concentration of not more than 0.005 M.

GENERAL COMMENTS OF PROCEDURE

The procedures described above are those which have been found most satisfactory and convenient by the authors. They are reproduced here as a guide only and can of course be modified in some degree to suit individual requirements. In this respect the comparative insensitivity of the quantum efficiency to wide changes in solution composition, light intensity and quality and total radiation dose is a particularly attractive feature of the ferrioxalate actinometer.

For the measurement of very low light intensities (e.g. for monochromatic light isolated by a small monochromator) a modification of the above procedure has been found useful. This involves the irradiation of 3 ml. of actinometer liquid in a 1 cm spectrophotometer cell and subsequent reaction with 0.5 ml. of a solution which is 1.8 N in sodium acetate and 1.08 N in sulphuric acid and containing 0.1% w/v 1:10-phenanthroline monohydrate. Measurement is carried out in the same cell, and the limiting sensitivity (corresponding to a 'difference optical density' of 0.002) corresponds to 6×10^{-10} mole of ferrous iron. For laboratory measurements of medium and high intensities and for general photochemical experiments where it is not necessary to exclude air completely, a silica cell (35 $\times$ 45 mm and 15 mm optical depth) fitted with a loosely fitting Polythene lid has proved convenient. For the most accurate work provision may be made for stirring by a stream of oxygen-free nitrogen. However, the error introduced by using an aerated and unstirred

solution amounts to only a small percentage provided that absorption is complete throughout the irradiation. For field use (e.g. the measurement of sun or skylight dosage) a convenient container is an unplasticized polythene tube about 2 in. in diameter and 1 in. deep with a 1 in. diameter window of appropriate filter material sealed in one end.

For accurate work, particularly at low light intensities, all manipulation of ferrioxalate solution should be carried out in a dark room. Kodak OB safelights provide adequate and safe illumination, but even dim electric light soon causes appreciable decomposition of the solution.

COMPARISON OF SENSITIVITY WITH THAT OF THE URANYL OXALATE ACTINOMETER

The quantum yield of the ferrioxalate solution is of the order of unity for all wavelengths up to $510\text{ m}\mu$ where light absorption per 5 cm depth is still 35 % (see figure 1). In contrast, the quantum yield of the uranyl oxalate solution has fallen to zero at $494\text{ m}\mu$ (Leighton & Forbes 1930) and the light absorption is 35 % per 5 cm at about $475\text{ m}\mu$ (see figure 2). Comparing the two actinometers at a wavelength of about

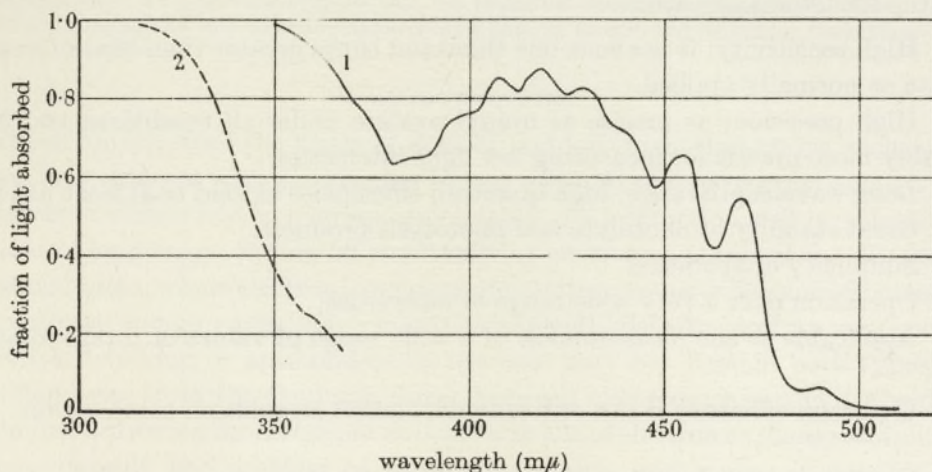


FIGURE 2. Light absorption by 'standard' uranyl oxalate actinometer solutions. Curve 1, 0.01 M-uranyl sulphate in 0.05 M-oxalic acid (50 mm optical depth). Curve 2, 0.001 M-uranyl sulphate in 0.005 M-oxalic acid (50 mm optical depth).

$350\text{ m}\mu$ we find that with 3 ml. of ferrioxalate solution, used as described above, $5 \times 10^{-10}\text{ E}$ of radiation can be detected. For the same volume of uranyl oxalate solution (0.05 M-oxalate) using the most careful weight titration method, a sensitivity one thousand times less cannot be exceeded unless more dilute solutions and shorter wavelengths are used.

Recently, a more sensitive method of using the uranyl oxalate actinometer has been described (Pitts, Margerum, Taylor & Brim 1955). This uses the principle of differential absorption spectrophotometry after reaction of the residual oxalate with a slight excess of ceric sulphate. For high sensitivity the method requires careful weighing of all solutions (four weighings for each determination), and the sensitivity will generally be governed by the accuracy of weighing rather than

by the accuracy of the spectrophotometric measurement. Thus, assuming the normal analytical weighing accuracy of ± 0.0001 g, a difference in weight of 0.0004 g between 'blank' and test solution is possible (errors in weighing the ceric sulphate solution being neglected). The minimum significant difference between the irradiated and non-irradiated solution will then be 4.0×10^{-8} mole of oxalate (corresponding to 0.0008 g of solution). At a quantum efficiency of 0.55 this would require 7.3×10^{-8} E. This is about 100 times less sensitive than the ferrioxalate actinometer. Again this could be reduced by using more dilute solutions but only at the expense of wavelength range.

It may be concluded that the ferrioxalate actinometer is more sensitive than all methods which have so far been devised for using the uranyl oxalate actinometer. It is also far more convenient.

CONCLUSIONS

Potassium ferrioxalate forms an ideal general purpose chemical actinometer for use over a wide range of wavelengths, and for many purposes it is superior to uranyl oxalate. Its advantages are:

- (a) High sensitivity; it is about one thousand times greater than that of uranyl oxalate as normally applied.
- (b) High precision; as precise as uranyl oxalate under all conditions and considerably more precise for measuring low light intensities.
- (c) Long wavelength range; high quantum efficiencies extend to at least $510\text{ m}\mu$.
- (d) Great stability of photolyte and photolysis products.
- (e) Simplicity of operation.
- (f) Operation over a very wide range of intensities.
- (g) Applicable to the measurement of a wide range of values of total radiation dose.
- (h) Quantum efficiency is not critically dependent on solution composition.

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